Sama Tawfik

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Summary

AI Engineering student with hands-on experience in Machine Learning and Computer Vision through academic projects and an ML internship. Passionate about building AI applications and currently expanding my knowledge in Large Language Models (LLMs) and Computer Vision.

Technical Skills

Programming Languages: Python, C++, SQL basics

Libraries & Frameworks: Pandas, NumPy,TensorFlow, Matplotlib,Scikit-learn,OpenCV,MediaPipe,YOLO

Tools & Platforms: Git & GitHub,Google Colab,VS Code

Soft Skills

Teamwork, Adaptability, Communication , Problem Solving, Fast Learner

Projects

Gaze Detection System Apr 2026 – May 2026
Python, OpenCV, MediaPipe
- Built a real-time gaze detection system using OpenCV and MediaPipe Face Mesh.
- Tracked facial landmarks to estimate eye gaze direction.
- Classified gaze into left, right, up, and down directions.
- Improved detection stability using geometric calculations.

Fraud Shield May 2026
Python, Scikit-learn, Pandas
- Built a machine learning model to detect fraudulent financial transactions.
- Performed data preprocessing and handled class imbalance.
- Trained and evaluated multiple classification models.
- Compared model performance using evaluation metrics.

Fashion-MNIST Classification May 2026
Python, TensorFlow, Keras, NumPy
- Built a neural network model to classify clothing images from the Fashion-MNIST dataset.
- Preprocessed and normalized image data before training.
- Trained and evaluated the model using TensorFlow/Keras.
- Analyzed model performance on unseen test data.

Handwritten Digit Classification May 2026
Python, TensorFlow, Keras, NumPy
- Developed a deep learning model to recognize handwritten digits using the MNIST dataset.
- Designed and trained a feed-forward neural network for multi-class image classification.
- Evaluated model performance using training and testing datasets.
- Applied deep learning concepts including activation functions and backpropagation.

TourPal Mar 2022 – Sep 2024
AI-Powered Travel Recommendation System
- Contributed to the development of an AI-powered travel recommendation platform.
- Participated in feature planning and system design to enhance user experience.
- Collaborated with the team to develop intelligent recommendation features.

Experience

Machine Learning Intern Sept 2025 – Nov 2025

Egypt Makes Electronics(EME)–ITIDA / Ideaspace

- Completed hands-on training in ML and AI concepts through a project-based learning initiative.
- Worked on practical ML tasks including data preprocessing, model training, and performance evaluation.
- Applied ML techniques to real-world datasets and projects in a collaborative environment.

Machine Learning Trainee March 2026 –June 2026

National Telecommunication Institute/NTI

- Implemented machine learning models using Python and Scikit-learn.
- Performed data preprocessing, feature engineering, and model evaluation.
- AWorked on hands-on assignments and practical machine learning projects.

Activities & Achievements

ISF Competition – 3rd Place Mar 2022-Sep 2024

A University-level competition

- Ranked 3rd place in the ISF competition through teamwork, problem-solving, and project presentation skills.

Education

Bachelor's Degree in Artificial Intelligence Badr University in Asyut (BUA) — Asyut, Egypt

Expected Graduation: 2027

- Third-year Artificial Intelligence student.
- Relevant coursework: Machine Learning, Deep Learning, Data Structures Algorithms, Computer Vision, Neural Networks, NLP

Certifications

- Egypt Makes Electronics (EME) — ITIDA / Ideaspace-ML
- Coursera-Deep Learning Specialization